

The 3GPP NB-IoT system architecture for the Internet of Things

Almudena Díaz Zayas* and Pedro Merino*

*University of Málaga, Andalucía Tech, Málaga, Spain, 29071

Email: almudiaz@lcc.uma.es, pedro@lcc.uma.es

Abstract—The first version of the 3GPP NarrowBand-IoT (NB-IoT) standards has been finalized in June 2016 as part of Release 13. NB-IoT is a promising new radio access technology which can coexist with existing GSM, UMTS and LTE deployments. In fact, NB-IoT specifications have been integrated into LTE standards. NB-IoT goes a step further than the MTC [3] (Machine Type Communication) specification, focusing on extremely low cost devices, massive deployments and reduced data rates with a carrier bandwidth of just 200 kHz (hence its name). In this paper we will take a close look at 3GPP specifications to discover which are the modifications required in traditional LTE deployments to provide connectivity to the upcoming Cat-NB1 User Equipments (UEs).

I. INTRODUCTION

As result of the 3GPP study item TR 45.820 [1] titled "Cellular system support for ultra-low complexity and low throughput Internet of Things (CIoT)" a new narrowband radio technology have been proposed, called NB-IoT. NB-IoT is being integrated in the LTE standards and a new UE category has been standardized in [2], Cat-NB1. Features provided by NB-IoT have been previously introduced in [4][5] and are oriented to cover the low-end massive Machine Type Communication (MTC) scenario: extended coverage, support of massive number of low throughput devices, high allowed latency, ultra-low device cost, low device power consumption and optimized network architecture. As shown in Table I the requirements of NB-IoT are the same as for eMTC (enhanced MTC)[3], but with focus on the low-end massive MTC scenarios. NB-IoT provides even lower cost in UEs than eMTC and a higher extended coverage of 164 dB MCL (at least for standalone deployment). The battery life is also around 10 years with 5 Watt Hour battery. NB-IoT is compatible with the enhanced DRX mode introduced in Release 13 to reduce power consumption. The number of devices supported per cell is around 55.000.

Unlike eMTC, that can only be deployed in-band, utilizing resource blocks within a normal LTE carrier, NB-IoT can also be deployed in the unused resource blocks within a LTE carrier's guard-band, or standalone for deployments in dedicated spectrum. NB-IoT is also particularly suitable for the re-farming of GSM channels.

An early proof of concept has been carried out by Aalto University and Ericsson Research in Finland. This initial tests were run in standalone mode and based on a Cloud RAN (C-RAN) architecture. The NB-IoT ability to support high

latencies make it an appropriate use case for applying a deployment based on C-RAN.

In this paper we describe the evolution of 3GPP standards related with MTC/NB-IoT and detail the changes introduced into LTE standards to provide support to NB-IoT, the latest 3GPP standard for IoT. The paper is organized as follows. In section II we provide the necessary background on 3GPP MTC/IoT standards to understand the role of NB-IoT in the IoT era. Section III describes the modifications introduced in the radio access network. The architectural transformation suffered in the EPC is explained in section IV. Finally, in section 5, we provide an additional comparison between MTC and NB-IoT, and introduce the future enhancements expected in Release 14.

II. BACKGROUND

3GPP (Third Generation Partnership Project) is the standardization body that specifies LTE/LTE-Advanced, as well as 3G UTRA and 2G GSM mobile communications systems. 3GPP standards are structured as Releases. The definition of the mobile technology known as Long Term Evolution (LTE) was initiated in 2005 and the first 3GPP specifications were introduced in Release 8 by in December 2007.

The term "user equipment" (UE) has been traditionally used by 3GPP to refer to the cellular equipment utilized by the subscriber to access network services. The UE can be a smart-phone or an embedded device contained in a M2M equipment. To support multiple UE hardware capabilities 3GPP defines different categories in 3GPP 36.306. The categories differ in the maximum data rates supported in the downlink and in the uplink, which are associated, for example, with the support of spatial multiplexing (MIMO transmission). The categories are also used by the eNodeB (LTE base station) to establish the conditions under which the communication with the UE will take place. As discussed below, the UE category is an important concept in IoT. Table II provides a summary of the UE categories oriented to MTC/IoT scenarios.

The initial version of the LTE Machine Type Communication (MTC) standards was introduced in Release 8 and was based on Category 1. Category 1 was the lowest capability with maximum bit rates of 10 Mbps downlink and 5 Mbps uplink while Category 5, the highest category in Release 8, supported data rates of 300 Mbps in downlink and 75 in Mbps in uplink. Although Category 1 UEs do not support MIMO transmission, they still have 2 receiver antennas and

	eMTC (Enhanced MTC) Release 13	NB-IoT (Release 13)
Coverage Enhancement	15 dB gain better compared to GPRS	20 dB gain compared to GPRS
Reduced complexity	Similar to EGPRS modem cost	Ultra-low cost (~1\$)
Power efficiency	10 year battery life	10 year battery life
Latency	10 sec	10 sec
Capacity	~10000 devices per cell	~50000 devices per cell
Coexistence	LTE in-band	GSM standalone, LTE in-band, LTE guard-band

TABLE I
eMTC VS NB-IoT FEATURES

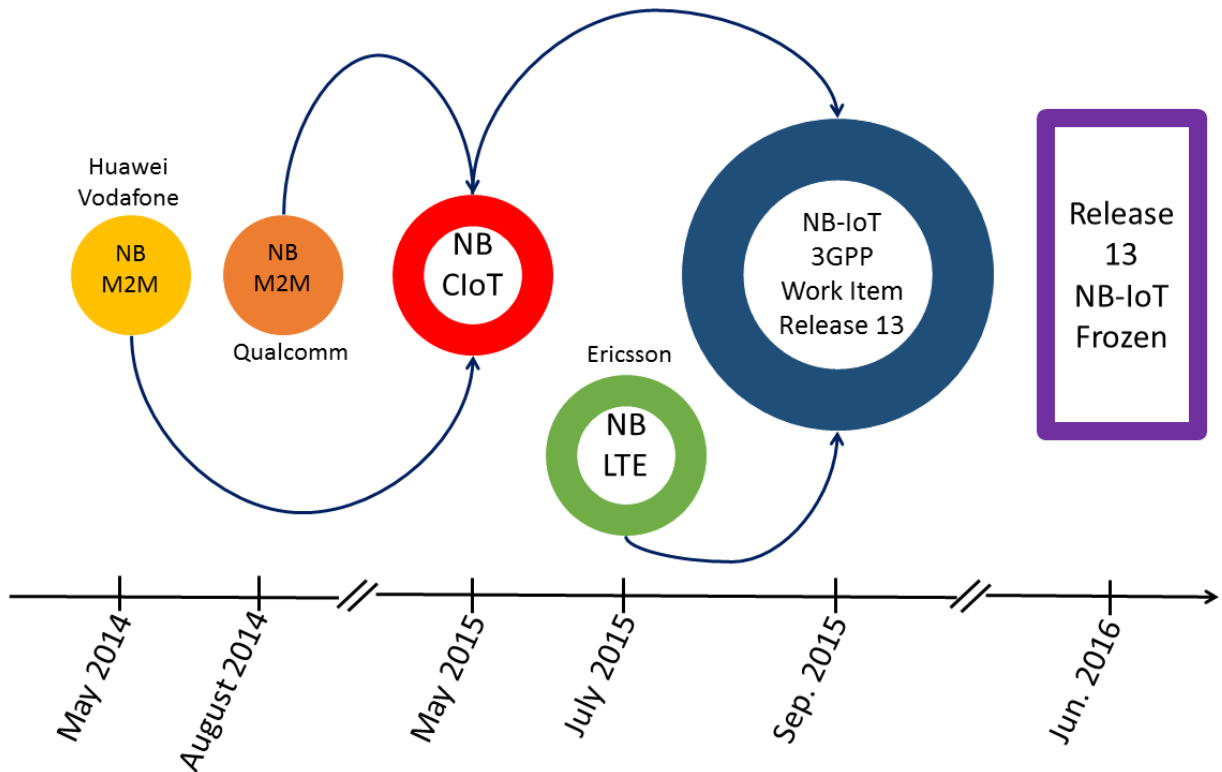


Fig. 1. NB-IoT standardization process

have to support all RF channel bandwidth options from 1.4 to 20 MHz and other functionalities that increase the hardware complexity. Moreover this category does not meet the battery, cost and range requirements of IoT. The first release of MTC was focused on the optimization of charging mechanisms, addressing, fixed location, low mobility and low activity terminals, handling of large numbers of subscriptions and subscriber data within the network, handling issues of large number of M2M subscriptions for the user of M2M services and security aspects.

A new UE category 0 was introduced in Release 12, which achieves cost reduction of about 50% compared to Category 1. Also a new power saving mode was introduced in this Release in 3GPP TS 24.301 and 23.682. The main purpose of this new feature is to reduce energy consumption while the device is not transmitting or receiving. Other solutions for MTC were discussed in Release 12 as part of 3GPP 36.888. Improvements such as half-duplex transmission, single RF chain and reduced peak throughput are part of the specification of MTC UE Cat 0.

Cost reduction using reduced system bandwidth and coverage improvements specification were postponed to Released 13.

Continuing the normative work started in Release 12 to enhance the LTE suitability for the promising IoT market, the key focus of eMTC (often referred to as LTE-M) for Release 13 is to define a new low complexity UE category type that supports reduced bandwidth, reduced transmit power, reduced support for downlink transmission modes, ultra-long battery life via power consumption reduction techniques and extended coverage operation. This is Cat-M1 (previously known as Cat-M). In Release 13 MTC achieves an additional 50% cost reduction through the restriction in system bandwidth to 1.4 MHz, which has a heavy impact on the design of the receiver, reducing the complexity of the baseband processing. This Release also introduces coverage enhancements to achieve an improvement of 15 dB, allowing operators to reach MTC devices in poor coverage conditions such as meters located in basements. These two improvements have an important impact on the design of the logical and the physical channels that

	Release-8	Release-12	Release-13	Release-13
	Cat. 1	Cat. 0	Cat. M1	Cat. NB1
Downlink peak	10 Mbps	1 Mbps	1 Mbps	200 kbps
Uplink peak rate	5 Mbps	1 Mbps	1 Mbps	144 kbps
Number of antennas	2	1	1	1
Duplex mode	Full duplex	Half duplex	Half duplex	Half duplex
UE receive bandwidth	20 MHz	20 Mhz	1.4 MHz	200 kHz
UE transmit power	23 dBm	23 dBm	20 dBm	23 dBm
Complexity	100 %	50 %	20 %	15 %
Use case	Voice services for emergency in elevators, Smart Grid Management, Kids/Elderly/Pet tracking	Cat0 is the interim solution prior Cat-M. Cat 0 is used for replacing Cat1, but cannot replace voice use cases.	Environment monitoring, Vehicle tracking	Smart metering, smart buildings, home automation
Availability	Available	Available	2017	2017

TABLE II
EVOLUTION OF UE CATEGORIES FOR MTC/IoT

have to support low bandwidth modes and enhanced coverage. Another key enhancement in Release 13 is the introduction of extended DRX [6], which is based on the usage of longer DRX (Discontinuous Reception) timers to achieve further power consumption improvements. The DRX concept consists in the monitoring of downlink signalling channel during a limited period of time. During the remaining time the device gets slept in order to reduce power consumption.

In May 2014, Huawei and Vodafone proposed a study item of NB-M2M to 3GPP GERAN which quickly garnered strong support and increased attention from other leading operators. In October of the same year Qualcomm submitted a new narrowband IoT proposal called NB-OFDM. In May 2015, both technologies merged into NB-CIoT (NarrowBand Cellular IoT).

Meanwhile Ericsson accelerated their research on narrow-band IoT and proposed NB-LTE (Narrowband LTE) in August 2015.

In September 2015, the 3GPP accepted the inclusion of both technologies as a Work Item in Release 13 [1]. NB-CIoT is a clean slate approach promoted by Huawei. The main difference between NB-LTE and NB-CIoT comes down to how much of existing LTE networks can be reused for IoT. NB-CIoT requires new chipsets and does not seem to be backwards compatible with any LTE network older than Release 13. NB-LTE, by contrast, can be fully integrated into existing LTE networks, works within current LTE bands and does not need an overlay network. In November 2015 3GPP agreed these two initiative evolve into only one standard called Narrowband IoT (NB-IoT). NB-IoT specifies a new radio access for cellular IoT, based to a great extent on a non-backward-compatible variant of E-UTRA, that addresses improved indoor coverage, support for massive number of low throughput devices, low delay sensitivity, ultra low device cost, low device power consumption and (optimised) network architecture. So, NB-IoT has become the main steam for 3GPP LPWAN (Low Power Wide Area Network) replacing NB-LTE and NB-CIoT (Cellular IoT) proposals. On June 2016 3GPP completed the standardization of NB-IoT, the new narrowband radio technology developed for the internet-of-Things (IoT).

NB-IoT UEs are referred to as Cat-NB1 (also known as Cat-M2). The reduction of their complexity, compared with Cat. 1, is up to 90%. 3GPP completed the standardization of NB-IoT on June 2016, it took only 9 months to standardize this new technology after the study phase, which demonstrates the importance of IoT for 3GPP.

The work carried out in GSM/EDGE group inside 3GPP on Extended Coverage GSM for IoT (EC-GSM-IoT) has been integrated into these two projects: LTE-M and NB-IoT.

In parallel with these standardization processes carried out inside 3GPP, there are other initiatives which operate in an unlicensed spectrum basis. It is the case of LoRaWAN, Sigfox and OnRamp wireless, Weightless -N & -P etc. Most of these networks take advantage of industrial, scientific, and medical ISM unlicensed frequency bands. These technologies are currently available, have already been deployed and meet the 4 factors for LPWAN (long range, very low power, low data rate, and very low cost). Some are based on protocols supported by industry alliances like LoRaWAN Alliance and Weightless SIG, some are based on proprietary protocols and some are standards in-progress.

III. NB-IoT RADIO ACCESS NETWORK

NB-IoT provides access to network services using a physical layer[8] optimized for very low power consumption and low cost. The full carrier bandwidth is 180 kHz, the subcarrier spacing can be 3.75 kHz (only for the uplink) or 15 kHz, the highest modulation scheme is QPSK, there is only support for FDD and half-duplex operation. The downlink of NB-IoT is based on OFDMA and the transmission scheme uses only one physical resource block (PRB). The uplink is based on single-carrier FDMA. For uplink transmission, there are two possible operation modes, single-tone transmission and multi-tone transmission. In single tone either 3.75 kHz or 15 kHz subcarrier spacing is allowed. In multi-tone only 15 kHz subcarrier spacing can be used. Multi-tone transmission enables grouping sets of 3, 6 or 12 subcarriers. Additionally, the minimum duration for the resource units used in scheduling depends on the number of assigned subcarriers and the oper-

ation mode, ranging from 1 ms in 12 sub-carriers multitone transmission to 32ms in 3.75 kHz single tone transmission.

A number of E-UTRA (Evolved Universal Terrestrial Radio Access) protocol functions supported by all Rel-8 UEs are not used for NB-IoT UEs: inter-RAT mobility, handover, measurement reports, public warning functions, GBR, CSG, support of HeNBs, relaying, carrier aggregation, dual connectivity, NAICS, MBMS, real-time services, interference avoidance for in-device coexistence, RAN assisted WLAN interworking, sidelink communication/discovery, MDT, emergency call and CS fallback.

For NB-IoT the positioning may be supported based on the existing LCS (Location Services) architecture with eNB measurements only.

Five new physical channels are introduced:

- Narrowband Physical Broadcast
- Narrowband Physical Downlink Shared Channel (NPDSCH). Carries the DL-SCH (Downlink Shared Channel) and PCH (Paging Channel) for NB-IoT UEs.
- Narrowband Physical Downlink Control Channel (NPDCCH). Informs the NB-IoT UE about the resource allocation of PCH and DL-SCH and carries the uplink scheduling grant for the NB-IoT UE.
- Narrowband Physical Uplink Shared Channel (NPUSCH). Carries the UL-SCH (Uplink Shared Channel) and Hybrid ARQ ACK/NACKs in response to downlink transmission for the NB-IoT UE.
- Narrowband Physical Random Access Channel (NPRACH). Carries the random access preamble for the NB-IoT UE.

Regarding MAC (Medium Access Control) layer, NB-IoT introduces a number of changes to reduce power consumption and make scheduling more flexible and simple. Because of the lower throughput requirements, a single HARQ process is used. This allows removing the HARQ identifier from the scheduling assignments and thus use a lower number of control bits for higher efficiency and robustness. Additionally, uplink retransmissions are no longer synchronous but always adaptive and asynchronous both in UL and DL. This provides the network with tighter control on the UL scheduling while preventing undesired periodic retransmissions as they will be now generated only when explicitly requested.

Other NB-IoT L2 and L3 enhancements include the reduction of the maximum PDCP SDU and Control PDU sizes, from 8188 octets to 1600 bytes. This is a reasonable tradeoff as traditional internet traffic does not exceed 1500 bytes per IP packet.

While in connected mode, DRX can be configured with cycles up to 10.24 seconds. Regarding Paging configuration, when DRX is enabled in idle mode, for NB-IoT, the maximum value of the DRX cycle is 10485.76 seconds (2.91 hours). This allows for very high power saving when the use case is compatible with accordingly high connection establishment latencies.

IV. NB-IoT SYSTEM ARCHITECTURE

NB-IoT architecture simplifies the existing EPC (Evolved Packet Core) architecture in order to address the IoT requirements of traffic models identified in [1] and shown in Table III. It bases small data transmission on a simplified architecture that is oriented to transporting these small data over NAS (Non Access Stratum) signalling messages. A dedicated core network node for CIoT (Internet of Things using 3GPP technology) profile provides combined C-plane and U-plane functions, e.g. aggregating some of the functions that traditionally reside in MME (Mobility Management Entity) and SGW (Serving Gateway, and in some instance PGW (Packet Data Network Gateway) in a new logical entity called C-SGN (CIoT Serving Gateway Node). C-SGN can be implemented to support only the necessary functionality required for CIoT use cases. S1-lite is an optimized version of S1-C (the S1 control plane which is based on the S1AP protocol) interface between the eNodeB and the MME. From S1-C only the necessary S1AP messages are supported (and inside the messages only the required parts) for the related CIoT procedures and the optimised security procedures are supported, as well as some functionality to support data transport. The user plane data is carried in these modified S1AP messages in order to support efficient small data handling. The data planes tunnels of the S1-U interface (defined in TS 23.401 [9]) are not required.

This architecture supports certain modifications (e.g. native support of SMS in PS domain, attach without PDN connection, simplified NAS etc.) to be applicable only to CIoT UEs without requiring procedures to maintain backwards compatibility with other UEs.

In NB-IoT, for signalling optimization, two solutions are introduced in addition to legacy RRC (Radio Resource Control) connection setup. The first solution, Data-over-NAS (DONAS), is mandatory and is a control plane optimization based on solution 2 proposed in [7]. DONAS enables transmitting data without having to activate an user plane, implementing only this option allows UE implementations with fewer requirements related to the user plane, such as the support for Unacknowledged Mode (UM) in RLC, multiple dedicated logical channels, DTCH, RLC re-establishment, DTCH...). The second solution is an user plane optimization called RRC Suspend/Resume. This mode is optional and is based on solution 18 specified in [7]). RRC Suspend/Resume introduces enhancements to disable and restore the user plane in an efficient way. In the control plane optimization the data is sent over Non Access Stratum (NAS), directly from the Mobility Management Entity (MME) in the core network to the UE without interaction from the base station.

The RRC Suspend/Resume procedures reduce the signalling overhead and also improve UE battery lifetime. The protocol architecture of NB-IoT and LTE is separated into control plane and user plane. Control plane consists of protocols which control the radio access bearers and the connection between the UE and the network. The highest layer of control plane is called Non-Access Stratum (NAS), which conveys

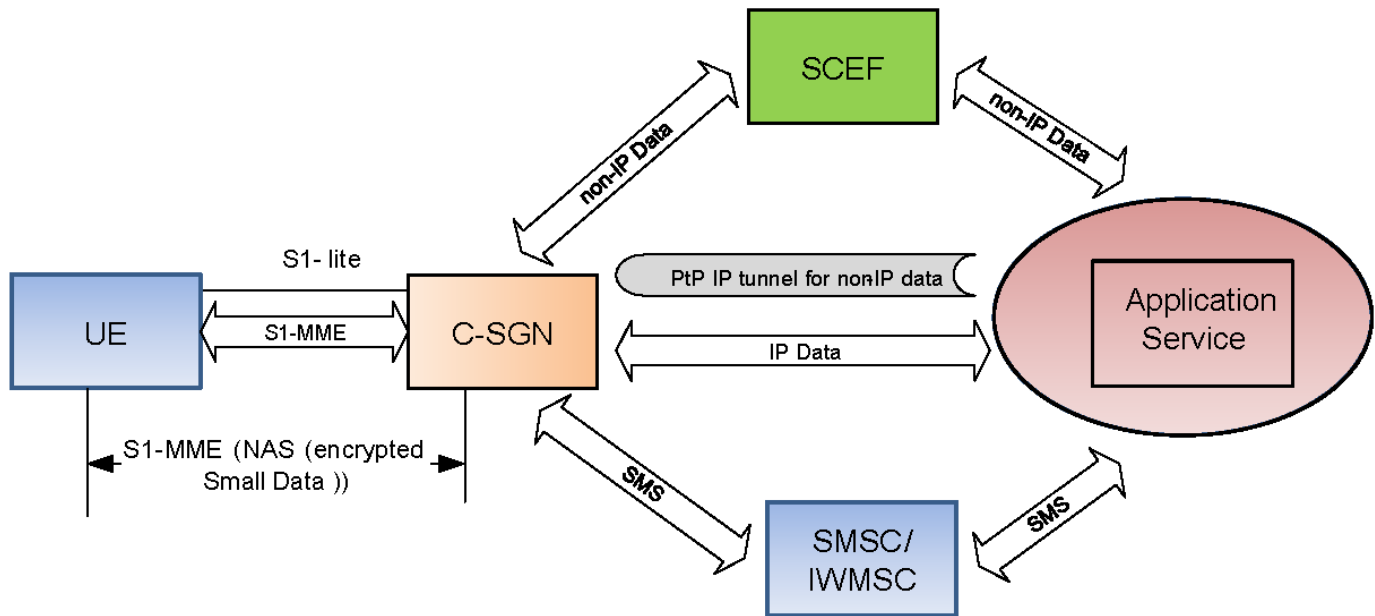


Fig. 2. LTE Architecture for NB-IoT

Category	Application example	UL Data Size	DL Data Size	Frequency
Mobile Autonomous Reporting (MAR) exception reports	smoke alarm detectors, power failure notifications from smart meters, tamper notifications etc.	20 bytes	0 ACK payload size is assumed to be 0 bytes	Every few months; Every year
Mobile Autonomous Reporting (MAR) periodic reports	smart utility (gas/water/electric) metering reports, smart agriculture, smart environment etc.	20 bytes with a cut off of 200 bytes i.e. payloads higher than 200 bytes are assumed to be 200 bytes.	50% of UL data size ACK payload size is assumed to be 0 bytes	1 day (40%), 2 hours (40%), 1 hour (15%), and 30 minutes (5%)
Network Command	Switch on/off, device trigger to send uplink report, request for meter reading	0 - 20 bytes 50% of cases require UL response.	20 bytes	1 day (40%), 2 hours (40%), 1 hour (15%), and 30 minutes (5%)
Software update/reconfiguration model	Software patches/updates	200 bytes with a cut off of 2000 bytes i.e. payload higher than 2000 bytes are assumed to be 2000 bytes.	200 bytes with a cut off of 2000 bytes i.e. payload higher than 2000 bytes are assumed to be 2000 bytes.	180 days

TABLE III
TRAFFIC MODELS FOR CELLULAR IoT [3GPP TR 45.820[1]]

the radio signalling between the UE and the EPC, passing transparently through the radio network. It is responsible for authentication, security control, mobility management and bearer management. Access Stratum (AS) is the functional layer below NAS, and in control plane it consists of Radio Resource Control protocol (RRC). RRC configures the user and control planes according to the network status. There are two main RRC states, RRC_Idle or RRC_Connected, and the RRC entity controls the switching between these states. In RRC_Idle, the network knows that the UE is present in the network and the UE can be reached in case of incoming call. In RRC_Connected the UE has a connection to the eNodeB, the network knows the UE location on cell level and the UE may receive and transmit data. Connection is released, switching to RRC_Idle, when there is no traffic to save the UE battery and radio resources. RRC Suspend/Resume procedures reduce the

signalling overhead required for UE state transition from Idle to Connected mode in order to have a user plane transaction with the network and back to Idle state by reducing the signalling messages required compared to legacy operation.

In order to support RRC Suspend/Resume procedures MME, eNodeB and UE have to include new functionalities. The MME, which is responsible for handling the mobility of the UE, has to support the new procedures on S1AP, in particular the S1AP procedure triggered by the "RRC Suspend" and "RRC Resume" procedures on the radio interface. The MME has also to store the UE Context information when the RRC connection is suspended and the S1AP association is kept. The eNodeB should include support for storing the UE context as well as the S1AP association related parameters when the UE is in the new CIoT RRC-Idle state. Moreover, the eNodeB has to support the new procedures to suspend/resume the

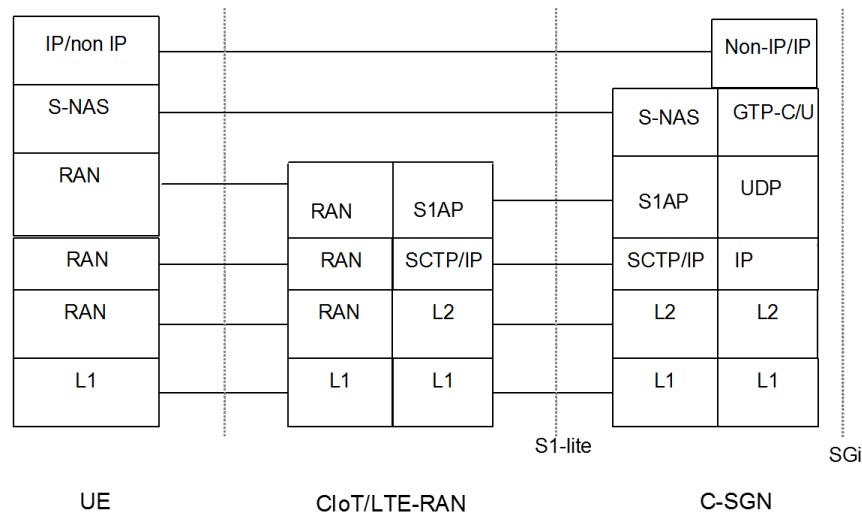


Fig. 3. Protocol stack for NB-IoT

connection. These procedures will be implemented between the eNB and the UE as well as between the eNB and the MME. Finally the UE has to store relevant AS information when the UE enters into the CIoT RRC_Idle and support the procedures needed to suspend and resume the RRC connection.

Data-over-NAS is considered as the basis for the normative work for support of infrequent small data transmission (for IP data, non-IP data and SMS). Support of DONAS is mandatory for both the UE and the network. This solution provides efficient support of infrequent small data transmission for IoT and support for non-IP data. The solution is based on lightweight core network architecture depicted in Figure 2. This solution piggybacks uplink small data onto the initial NAS uplink message that is extended and it uses an additional NAS message for carrying downlink response small data. The efforts for setting up the user plane, i.e. related RRC messages and AS security setup, can be avoided. There is not establishment of DRBs and S1-U bearers. Data can also be delivered through the SCEF (Service Capability Exposure Function), which is used for the delivery of non-IP data over control plane. SCEF element is part of the Architecture Enhancements for Service Capability Exposure (AESE) [10] introduced in Release 13, which displays network services to 3rd parties. The implementation of this solution also implies modifications of the procedures available at the UE, eNodeB and C-SGN/MME to support e.g. handling of "NAS small data" sessions and the attach without bearer activation.

The resulting protocol stack after the adoption of these two solutions is the simplified protocol stack shown in Figure 3 where S-NAS indicates a simplified NAS signalling.

V. CONCLUSION

While LTE-M is a pure LTE technology NB-IoT is a new radio technology that coexists with existing GSM, UMTS and LTE systems.

In regard to the uses cases of both technologies the main differences are mobility and throughput. NB-IoT offers reduced data rate and reduced mobility support compared to LTE-M as shown in Table II

In Release 14, we can expect further enhancements for NB-IoT in the form of multicast support, latency reduction optimizations to enable vehicle to vehicle (V2V) communications, positioning, mobility and service continuity enhancements, non-anchor PRB enhancements and new power classes and

ACKNOWLEDGMENT

This work has been funded by the Spanish Ministry of Economy and Competitiveness (TIN2015-67083-R), FEDER and the European Unions Horizon 2020 research and innovation programme (grant agreement No 688712).

REFERENCES

- [1] 3GPP TR 45.820 Cellular system support for ultra-low complexity and low throughput Internet of Things (CIoT)
- [2] 3GPP TS 36.101 Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) radio transmission and reception
- [3] 3GPP 36.888, Study on provision of low-cost Machine-Type Communications (MTC) UEs based on LTE
- [4] J. Gozalvez, "New 3GPP Standard for IoT [Mobile Radio]," in IEEE Vehicular Technology Magazine, vol. 11, no. 1, pp. 14-20, March 2016.
- [5] S. Landström, J. Bergström, E. Westerberg, D. Hammarwall, "NB-IoT: A sustainable technology for connecting billions of devices", in Ericsson Technology Review, vol. 93, no. 3, pp. 2-11, April 2016
- [6] 3GPP TR 23.770, Study on System Impacts of Extended DRX Cycle for Power Consumption Optimization
- [7] 3GPP TR 23.720 Study on architecture enhancements for Cellular Internet of Things
- [8] 3GPP TS 36.300 LTE; Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Universal Terrestrial Radio Access Network (E-UTRAN); Overall description; Stage 2
- [9] 3GPP TS 23.401 General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access
- [10] 3GPP TR 23.708 Architecture Enhancements for Service Capability Exposure (AESE)